

Supplementary Material for “Definition-Consistent Variable-Order Fractional Operators from Fixed-Pole Banks: Structure, Error Guarantees, and Order Tracking”

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Abstract

This supplement collects secondary numerical results and implementation details. Section and equation numbers without the prefix S refer to the main paper.

S1 Continuous-time switching results

Table S1 gives the numbers behind Fig. 2 of the main paper (Sect. 7.2).

S2 State maps for moving-pole realizations

This section expands Sect. 7.3 of the main paper.

Setup.

For six single switches (at sample 2000, window 2000), we fitted Φ by truncated least squares on a mixed training ensemble of 4 classes with 200 inputs each (white, piecewise-constant, multisine, Brownian). We tested on fresh ensembles of 150 inputs per class and on a unit step, sweeping the truncation threshold. Besides the most accurate map, we report a robust *minimum-norm* map: the map of smallest norm whose validation error is within twice the best. The maps were then rounded to float32 and to 24-bit fixed point.

Table S1 Continuous time, step input: relative RMS error after a single switch. FP-out and FP-in are the fixed-pole bank with output and input schedule; Ou-par and Ou-cas are the hot-swapped Oustaloup forms. Arrows show $S = 15 \rightarrow 41$; the cascade column is at $S = 41$

$\alpha_1 \rightarrow \alpha_2$	A-B gap	FP-out vs A	FP-in vs B	Ou-par vs A	Ou-cas, A / B
$-0.3 \rightarrow -0.7$	0.52	$1.0\text{e-}4 \rightarrow 1.7\text{e-}4$	$1.0\text{e-}4 \rightarrow 5.1\text{e-}5$	$0.12 \rightarrow 0.042$	0.16 / 0.33
$+0.3 \rightarrow +0.7$	0.65	$2.1\text{e-}3 \rightarrow 1.5\text{e-}5$	$4.2\text{e-}3 \rightarrow 1.5\text{e-}4$	$0.14 \rightarrow 0.053$	1.6 / 1.7
$-0.5 \rightarrow +0.5$	0.81	$1.1\text{e-}3 \rightarrow 4.3\text{e-}5$	$5.1\text{e-}4 \rightarrow 3.1\text{e-}5$	$0.23 \rightarrow 0.085$	4.6 / 1.6
$+0.5 \rightarrow -0.5$	1.35	$9.8\text{e-}5 \rightarrow 3.4\text{e-}5$	$9.4\text{e-}4 \rightarrow 8.0\text{e-}6$	$0.25 \rightarrow 0.080$	0.24 / 0.87

On the fixed band $[10^{-3}, 10^5]$ rad/s used here, the fixed-pole bank reaches its tail floor near $S = 15$. The rule of Sect. 5 chooses the band from the tolerance instead.

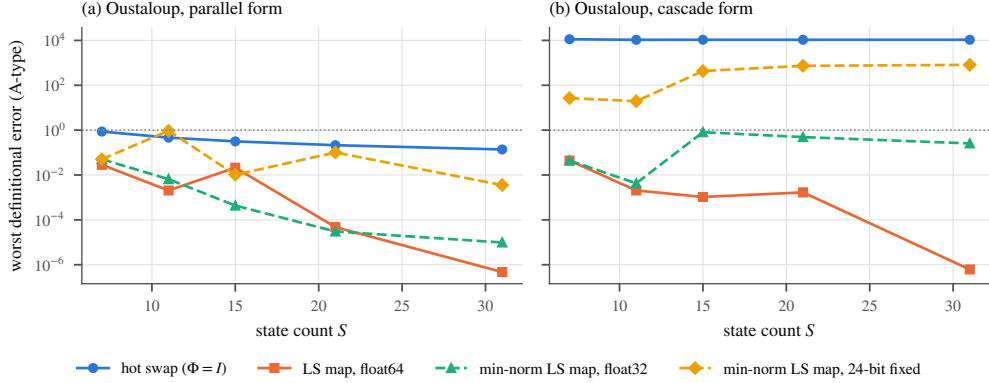


Fig. S1 Worst definitional error (A-type) after one switch, over six switches and five input classes including the step. Hot swap versus least-squares state maps in float64, and the minimum-norm map rounded to float32 and to 24-bit fixed point. The dotted line marks an error of 1

Results.

Figure S1 and Table S2 report the worst case over the switches and the input classes. In float64 the error falls from 0.14 (parallel, hot swap) and 1.1×10^4 (cascade, hot swap) to 1.7×10^{-7} and 6.2×10^{-7} at $S = 31$. The map is, however, a dense $S \times S$ matrix, with norm between 10^3 and 5×10^4 even for the minimum-norm choice. Rounded to 24 bits, the map loses 3.5 to 4.3 orders of magnitude of accuracy in the parallel form and 7.4 to 9.1 in the cascade form for $S \geq 21$. In the cascade form the 24-bit error exceeds 1, and even float32 rounding raises it to 0.26–0.81 for $S \geq 15$.

With a smooth derivative order quantized to 41 levels (80 grid crossings, parallel form), a chain of maps reduced the step-input error from 4.1×10^{-2} to 1.2×10^{-4} ($S = 15$). The fixed-pole bank needs no map, and its definitional error is exactly zero. For the $P = 1024$ order levels of Sect. 9, maps between neighbouring levels alone would need about 56 Mbit of storage at $S = 33$ and 25 bits per entry, against 0.78–1.08 Mbit for the whole coefficient table of the bank.

Table S2 State maps for the Oustaloup forms: worst A-type definitional error over 6 switches and 5 input classes (including the step), and the largest norm of the minimum-norm map

S	form	hot swap	LS, float64	minimum-norm map		
				float64 ($\ \Phi\ _2$)	float32	24-bit
7	parallel	0.86	2.8e−2	5.0e−2 (3e4)	5.0e−2	5.0e−2
15	parallel	0.32	2.1e−2	4.4e−4 (5e4)	4.3e−4	1.0e−2
31	parallel	0.14	4.7e−7	1.7e−7 (5e3)	9.8e−6	3.6e−3
7	cascade	1.1e4	4.4e−2	4.4e−2 (9e2)	4.4e−2	2.7e1
15	cascade	1.1e4	1.0e−3	3.2e−4 (8e3)	8.1e−1	4.3e2
31	cascade	1.1e4	6.2e−7	6.2e−7 (4e4)	2.6e−1	8.1e2

The fixed-point test rounds Φ to a single Q format; per-row scaling might do better. The fixed-pole bank needs $\Phi = I$ and has zero definitional error.

S3 Which type does a hot swap implement?

Using the own-type references of Sect. 7, we asked which of the four types each hot-swapped realization is closest to, over 8 cases (four piecewise-constant order profiles, step and white-noise inputs) for $N = 7$ and $N = 15$:

- the fixed-pole bank is closest to A (output schedule) or B (input schedule) in all cases, at distances $\leq 2.1 \times 10^{-14}$;
- the parallel Oustaloup form is closest to the A-type in all cases, at distances up to 0.42;
- for the cascade form with $\omega_h = 10^3$ rad/s, the closest type is A, B, D and E in two cases each, so the cascade has no consistent type; with $\omega_h = 10^5$ rad/s it is A or B, at distances up to 1.3×10^3 ;
- with $\omega_h = 10^5$ rad/s, above the Nyquist frequency, the zero-order-hold members of the integral approximations have zeros outside the unit circle (moduli 3 to 41), and their own D- and E-types cannot even be formed in 6 of 8 cases.

S4 Fixed-point realization: details

This section expands Sect. 9 of the main paper. The design point is $R = 4000$, $\varepsilon = 10^{-4}$ and $|\alpha| \leq 0.95$, giving $K = 32$ plus the delay mode, and the orders come from a grid of $P = 1024$ levels, $\alpha_i = 0.95(2i/(P-1) - 1)$.

Arithmetic in sample units

The core runs in sample units ($T_s = 1$), where $\hat{g}_0 = 1$ for every order. The inversion (21) then needs no divider: $v_n = x_n - \eta_n$. Physical units follow from one gain $T_s^{-\alpha_n}$, applied at the output for the A- and E-types and at the input for the B- and D-types. This follows directly from (2)–(5); for the E-type, substitute $z_n = T_s^{-\alpha_n} \zeta_n$.

The number formats are as follows:

- *Signals*: W_S -bit two’s complement with F fractional bits.

- *States*: $W_{ST} = W_S + 13$ bits. Each mode stores its state scaled by a power of two 2^{G_k} (one gain per schedule) so that its own bound uses the full word. Without this scaling, the input-scheduled slow modes lose their increments $c_k v_n$, which lie far below the signal LSB; the B- and E-type errors were then 2×10^{-2} to 0.66.
- *Coefficients*: a W_M -bit mantissa with a 7-bit shift.

Products are rounded as $\text{mulsh}(a, M, S) = (aM + 2^{S-1}) \gg S$. Algorithm 1 of the main paper gives one sample of the core in exact arithmetic; the bit-accurate golden model and the RTL use the same sequence with the formats above.

Dynamic range: two operating envelopes

For $|x| \leq 1$ over 4000 samples, the frozen-order ℓ_1 bound on the output is 2696. For the A-type this bound holds under any switching, because each output uses the weights of a single order; for the B-type the switching stress tests stayed within it. For the D- and E-types, switching from an integral to a derivative order drives the output to 2.2×10^6 under a DC input, 820 times the frozen-order bound. The new inverse acts on a history that has grown under the old order. This is the behaviour of the recursive definitions themselves, not an artefact of the bank. We therefore size two envelopes with a factor-two margin: 14 integer bits for the forward types only (A/B), and 24 integer bits for all four types (A/B/D/E).

Quantization and inverse stability

Quantization can undo the DC floor. Coefficients and pole steps are dyadic rationals, so the signs of $\hat{H}_q(\pm 1)$ of every quantized ROM row can be evaluated exactly in rational arithmetic, which we did. With the floating-point floor alone:

- in the A/B configuration, 64 of the 512 positive-order rows have $\hat{H}_q(1) \leq 0$, the worst having its dominant inverse pole at $1 + 4.8 \times 10^{-6}$ (e-folding time 2.1×10^5 samples);
- in the A/B/D/E configuration, 51 of 512 rows have $\hat{H}_q(1) \leq 0$, the worst ($\alpha = 0.855$) having its pole at $1 + 1.9 \times 10^{-9}$ (e-folding time 5.2×10^8 samples, about 9 minutes at 1 MS/s).

$\hat{H}_q(-1)$ was positive in every row. The *quantization-aware* DC floor raises the delay mantissa in LSB steps until $\hat{H}_q(1)$ is at least the floating-point value $\hat{H}(1)$. It brings both counts to zero without any measurable change in accuracy.

Word length

We swept $W_S \in \{32, \dots, 56\}$ and $W_M \in \{16, 18, 25, 32\}$ (Fig. S2). The test set comprised 3 order profiles (random piecewise, a new order every sample, a sign-changing switch), 3 inputs, and all types of the envelope. A configuration was accepted if, for every type, its error against the literal definition was at most the floating-point bank's error plus 0.1ϵ . The smallest accepted configurations were:

- A/B: $W_S/W_{ST}/W_M = 36/49/16$;
- A/B/D/E: $48/61/25$.

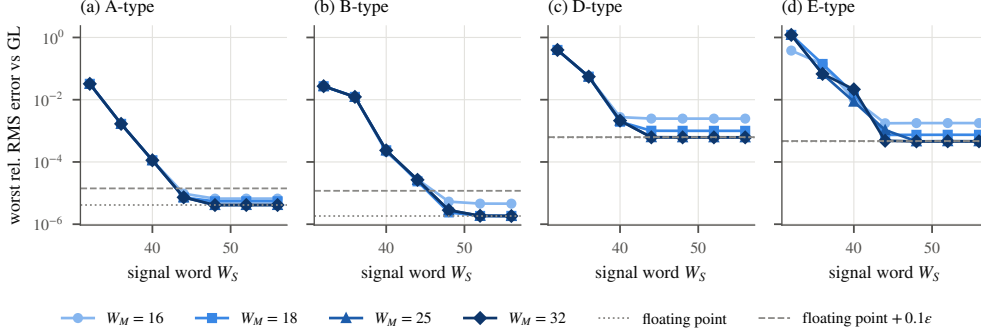


Fig. S2 Word length, A/B/D/E envelope: worst relative RMS error against the literal definitions versus signal word W_S (state word $W_S + 13$), for four mantissa widths. Dotted: floating-point bank; dashed: acceptance limit

With an 18-bit mantissa the D- and E-types stall near 10^{-3} , because the inversion is ill-conditioned: the low-frequency gain of a derivative bank is about $R^{-\alpha}$. The floating-point bank itself has D- and E-type output errors up to 6×10^{-4} at this design point, although its weight error is 10^{-4} . This is the cancellation effect of Corollary 3, amplified by the inversion.

S5 Streaming mBm: details

This section expands Sect. 8.3 of the main paper.

Further alternatives to mBm treat a varying Hurst exponent differently. The incremental mBm of Ślęzak and Metzler [1] is defined through the covariance of its increments and shares the property that a change of H acts only on new increments. Other alternatives to mBm include:

- the multifractional processes of Surgailis, built by nonhomogeneous fractional integration of white noise [2], whose covariance is not a local function of $H(t)$ and $H(t')$ as it is for mBm [3];
- the integrated fractional white noise of Sly, proposed to avoid the extreme changes in magnitude of mBm [4].

Conversely, the variable-Hurst fBm of Ryvkina uses the current Hurst value in its kernel and has discontinuous paths wherever H jumps [5], like the A-type. VO operators have been used to synthesize multifractional Gaussian noise before [6], and simulation methods for mBm include [7]. Exact methods for stationary Gaussian processes such as circulant embedding [8] rely on stationarity and do not apply directly when H varies.

Results.

Table S3 and Fig. 8 summarize the results. The bank was designed by the rule for $\epsilon = 10^{-4}$ and $0.05 \leq H \leq 0.95$, giving $K = 22, 24$ and 26 for $n = 1024, 4096$ and 16384 .

Table S3 Streaming mBm with the fixed-pole bank

Quantity	Result
integer split, correct order (A first, B last)	$\leq 9 \times 10^{-15}$
integer split, reversed order	0.1 to 1.3
path vs literal GL, same noise	$A \leq 1.9 \times 10^{-6}$, $B \leq 2.4 \times 10^{-6}$
exact variance / covariance ($n = 2048$)	$\leq 3.1 \times 10^{-5}$ / $\leq 1.7 \times 10^{-6}$ (Frobenius)
local Hurst tracking, varying H (400 paths)	RMSE of mean 0.014 to 0.036, A and B alike
variance before/after a jump of H from 0.3 to 0.8	A $119 \rightarrow 1.5 \times 10^5$; B $119 \rightarrow 119$
cost per sample	bank 47 to 55 MACs (n -independent); direct GL $n/2$ on average
fixed-point core A/B (36/49/16), vs float	A 1.8×10^{-5} , B 8.9×10^{-4}
fixed-point core A/B/D/E (48/61/25), vs float	A 3.8×10^{-6} , B 1.2×10^{-4}

Second-order statistics were computed exactly, without Monte Carlo, from the $n \times n$ weight matrix of the generator. The variance agreed with the exact formula to 3.1×10^{-5} and the covariance to 1.7×10^{-6} .

The local Hurst exponent was estimated by generalized quadratic variations [9, 10]; see also [3]. For constant H the estimator itself has a bias of +0.06 at $H = 0.2$ and a standard deviation of 0.06 (window 1024). For varying H its mean followed $H(t)$ with RMSE 0.014 to 0.036, equally for both types.

The two types differ sharply at a jump of H from 0.3 to 0.8. The A-type variance jumps from 119 to 1.5×10^5 , and the path is discontinuous: the squared increment at the jump is 1.3×10^5 times the typical one. The B-type is continuous (ratio 1.0), in line with the behaviour reported for MMFBM and incremental mBm [1, 11]. Which definition to use is therefore a modelling decision with visible consequences, as also argued in [11]. The bank makes either choice available at the same cost.

Hardware.

The cores of Sect. 9 generate the same processes bit-exactly from integer noise. For the A-type, the accumulator sits in front of the core and is exact (an integer sum), so the small A/B core suffices. For the B-type, the accumulator follows the core and integrates its low-frequency rounding errors. The B-type therefore benefits from the wider configuration (Table S3, last two rows). Input scales were 2^{-9} for A and 0.25 for B, chosen to keep the accumulated A-type input within the signal range.

S6 B-type step response

The B-type operator in continuous time processes each input sample with the kernel of its entry order: $y(t) = \int_0^t K(t - \tau, \alpha(\tau))u(\tau) d\tau$ with $K(s, a) = \partial_s F(s, a) =$

$s^{-a-1}/\Gamma(-a)$, taken in the finite-part sense for $a > 0$. For $u = 1$, let $G(\tau) = F(t - \tau, \alpha(\tau))$. Then

$$\frac{dG}{d\tau} = -\partial_s F(t - \tau, \alpha(\tau)) + \partial_a F(t - \tau, \alpha(\tau)) \dot{\alpha}(\tau),$$

so $y(t) = G(0) - G(t) + \int_0^t \partial_a F \dot{\alpha} d\tau$. The finite part removes $G(t) = F(0^+, \alpha(t))$, and $G(0) = F(t, \alpha(0))$. This gives (9). For a piecewise-constant order, $\dot{\alpha}$ is a sum of Dirac impulses and the integral becomes the stated finite sum. In the experiments the smooth case was evaluated by adaptive quadrature after the substitution $s = v^m$ with $m(1 - \alpha_{\max}) \geq 2$, which removes the weak singularity at $s = 0$.

S7 Large-lag alias term

For large r the integrand of (14) is concentrated at $u = O(1/r)$, where $(1 - e^{-u})^a \approx u^a$. Hence $f(x) \approx e^{(1+a)x} e^{-\rho e^x}$ with $\rho = r - a$. Its Fourier transform is $\hat{f}(\omega) = \Gamma(1 + a - i\omega) \rho^{-(1+a-i\omega)}$, and $\hat{f}(0) = \Gamma(1 + a) \rho^{-(1+a)}$ is the integral itself. The relative error of the infinite trapezoidal rule is $\sum_{k \neq 0} \hat{f}(2\pi k/h) / \hat{f}(0)$, dominated by $k = \pm 1$. Stirling's formula, $|\Gamma(1 + a - i\omega)| \sim \sqrt{2\pi} \omega^{a+1/2} e^{-\pi\omega/2}$, at $\omega = 2\pi/h$ gives (23). The result does not depend on r , which justifies the name large-lag term.

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